

# DR JOSEPH MANGHAN

SENIOR FRONTEND ENGINEER | CONTEXT ENGINEERING

 Cardiff, Wales |  josephmanghan@gmail.com |  [www.josephmanghan.com](http://www.josephmanghan.com) |  [in/josephmanghan](https://in.josephmanghan) |  [github/josephmanghan](https://github.com/josephmanghan)

Capable, committed and enthusiastic Senior Frontend Engineer with specialization in context engineering and spec-driven development. Experienced in implementing structured AI-first workflows to enhance team productivity and code quality. Passionate about UX, responsive design, and iterative improvement. Expertise across the full SDLC.

## Experience

### Senior Frontend Engineer

*Backbase | Jan 2026 - present*

**Support and empower my squad in delivering essential features within the Identity Access Management space, primarily focusing on auth flows for banking contexts.**

Led the technical design for complex IAM features, including orchestrating input from stakeholders such as UX, BE and Tech Writers.

Well-versed in harnessing AI to expedite development, learning and support

- Formally seconded to company-wide task force delivering a context engineering and spec-driven development strategy — centralised skills, evaluations, and versioned assets, with an upskilling stream covering models, tooling, and harnesses.
- Reduced testing overhead by introducing dedicated review and QA agent personas for the post-development phase.
- Directed workshops on context engineering and AI, facilitating discourse on innovative workflows and product development.

Oversaw the organisation, story mapping and UX liaison of a major refactor initiative employing a shared componentisation strategy

- Resulted in a significant improvement to code quality and UX parity across the product

## **Frontend Engineer**

*Backbase | Jan 2023 - Dec 2025*

**Consolidated frontend expertise through hands-on delivery, quality advocacy, and iterative improvement in IAM feature development.**

Managed prompt assets, context, and specs at squad level to support day-to-day feature delivery.

## **Junior Frontend Engineer**

*Backbase | Aug 2021 - Dec 2022*

## **Web Development Internship**

*Rhys Welsh Web Design | Jun - Jul 2021*

**Foundational experience with CMS content architecture.**

## **PhD in Web Science & Music**

*University of Southampton | 2015 - 2020*

**Utilised web practices to outsource compositional tasks to crowds.**

## Projects

|                                                                                                                                                                |                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="http://www.josephmanghan.com">www.josephmanghan.com</a><br><br>Designed, coded, and deployed via a single prompt - a study in long-horizon prompting. | <a href="#">skill-style-guide</a>   Published to <a href="https://skills.sh">skills.sh</a><br><br>Style conventions for agent skills and prompts - making prompt review as systematic as code review. |
| <a href="#">devai-cli</a><br><br>Local-first CLI tool for AI-powered git commit message generation using Ollama.                                               | <a href="#">gem-factory</a><br><br>Development environment for deploying Gemini Gems with complex workflows.                                                                                          |

|  Skills                                   |  Education |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Technical:</b><br><br>Typescript • Angular • React • Jest • Vite • Playwright • Cypress • SCSS                          | <b>Pitman Training   The Tech Academy</b><br>Web Developer Diploma (Dist.)<br><br>2020-21   |
| <b>AI Tooling:</b><br><br>Context Engineering • Cursor • Claude Code • Gems • Spec-driven development • Prompt engineering | <b>University of Southampton</b><br>PhD in Web Science & Music<br><br>2015-20               |
| <b>Professional:</b><br><br>Strong written & verbal communication • Mentorship • Public speaking • Research & analysis     | MMus Music Composition (Dist.)<br>2014-15<br>BA Music (1st class)<br><br>2011-14            |